Google Cloud H2S

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.

Participant Details

a. Participant name: Loga Rajeshwaran K
b. Problem Statement: Professionals and students waste hours manually reading large texts — there is no quick, intelligent way to summarize, extract key points, and classify content automatically.

Brief about the idea

The **Smart Summarizer Agent** is an AI-powered text intelligence system that instantly processes large volumes of text, delivering summaries, key points, and classifications in seconds.

What it does:

- Accepts any text input and generates a concise summary, headline, and 3 key takeaways
- Classifies the topic into categories like Technology, Business, Health, and more

How it works:

- Built using **Google ADK** for agent orchestration and **OpenAI GPT-4o-mini** via LiteLLM as the language model
- Exposed as a **REST API** with FastAPI and deployed on **Google Cloud Run** for serverless, auto-scaling hosting

Why it matters:

- Saves hours of manual reading by delivering instant, structured insights from any text
- Fully API-driven and scalable — easily plugs into any existing workflow or application

Meeting the Build Criteria

- How did you navigate the requirements of your chosen track to build a custom solution - using ADK, MCP, or AlloyDB AI that effectively addresses the 'What You Must Build' criteria?

Chose the ADK Track — Built the Smart Summarizer Agent using Google ADK (Agent Development Kit), which provides native agent orchestration, tool registration, session management, and async runner — all the core requirements of the ADK track
Custom Tool Design — Developed two purpose-built ADK tools (summarize_text and classify_text) that the agent intelligently selects and calls based on the user's request, demonstrating real agentic decision-making
LLM Integration via LiteLLM — Connected OpenAI GPT-4o-mini as the language model through ADK's LiteLlm wrapper, making the agent model-agnostic and production-ready
REST API Layer — Wrapped the ADK agent in a FastAPI server with structured request/response models, making it accessible to any client or downstream system
Cloud Run Deployment — Deployed the full solution on Google Cloud Run as a containerized, serverless service — meeting the scalability and production-readiness criteria

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

Unlike ChatGPT or Notion AI, Smart Summarizer Agent is a fully API-driven service built on Google ADK with real agentic tool-calling — not just a prompt wrapper.

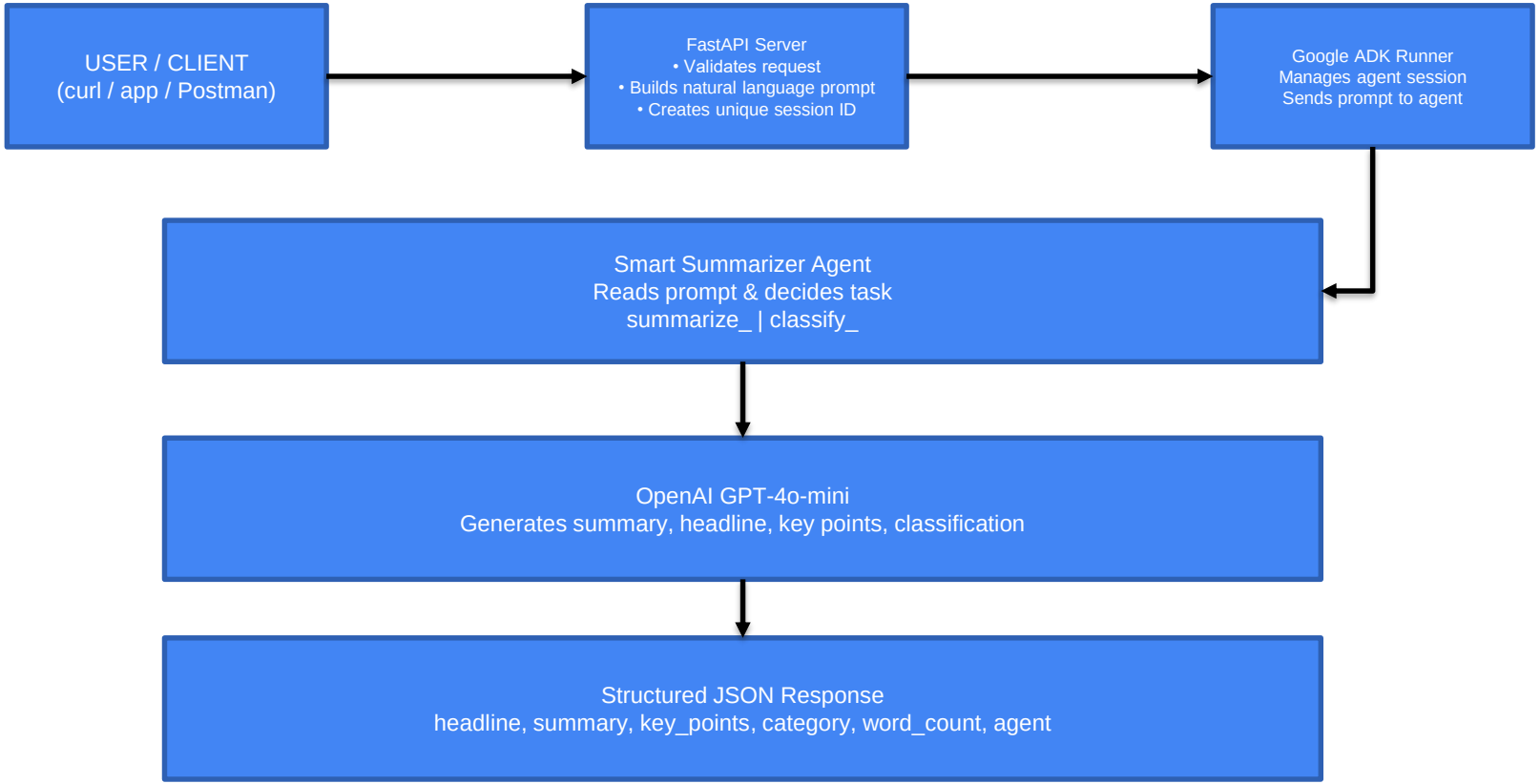
A single API call returns a summary, headline, key points, and topic classification instantly — eliminating manual reading and deployable at any scale via Google Cloud Run.

Multi-capability in one call + model-agnostic architecture via LiteLLM — the only free, embeddable API that combines summarization, classification, and headline generation in a single serverless endpoint.

List of features offered by the solution

- Text Summarization — Converts long text into a short summary in 3 styles: concise, detailed, or bullet points
- Headline Generation — Produces a single punchy title that captures the essence of the text
- Key Point Extraction — Automatically pulls out the 3 most important takeaways from any text
- Topic Classification — Categorizes text into domains like Technology, Business, Health, Politics, and more
- Multi-task in One Call — Summarize and classify simultaneously using the task: "both" option
- REST API Interface — Simple JSON request/response, integrates with any app or workflow
- Style Flexibility — Users choose their preferred summary style per request
- Serverless Deployment — Hosted on Google Cloud Run, auto-scales with zero infrastructure management
- Model-Agnostic — Swap underlying LLM (OpenAI, Gemini, etc.) without changing any code
- Interactive API Docs — Built-in Swagger UI at /docs for easy testing and exploration

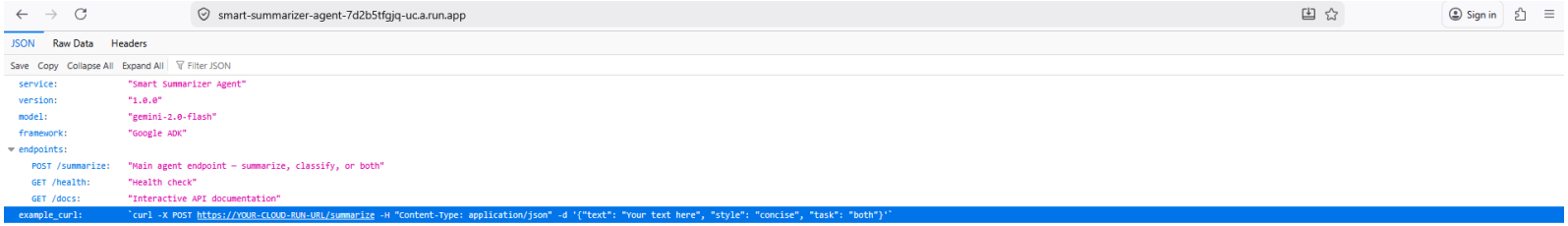
Process flow diagram or Use-case diagram

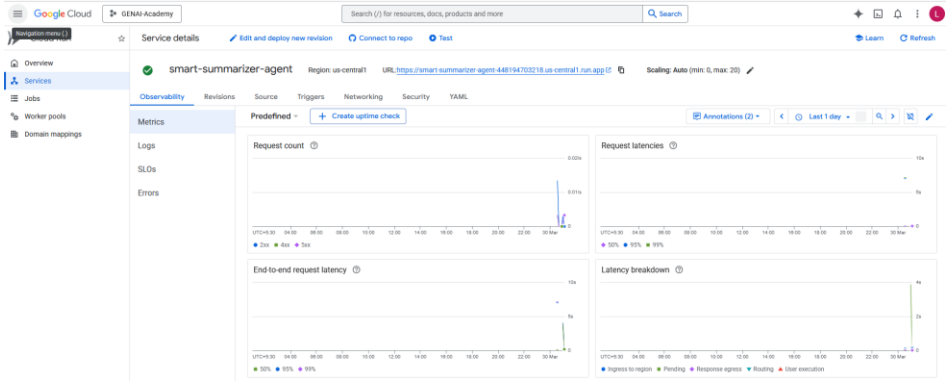


Technologies to be used in the solution

Layer	Technology	Purpose
Agent Framework	Google ADK	Agent orchestration, tool-calling, session management
Language Model	OpenAI GPT-4o-mini	Text summarization, classification, headline generation
LLM Bridge	LiteLLM	Model-agnostic LLM integration layer
API Framework	FastAPI	REST API server with request validation and routing
Server	Uvicorn	ASGI server to run FastAPI
Containerization	Docker	Package app and dependencies into a portable image
Cloud Hosting	Google Cloud Run	Serverless, auto-scaling deployment
Language	Python 3.11	Core programming language
Data Validation	Pydantic	Request/response schema validation
Config Management	python-dotenv	Manage API keys via .env file

Snapshots of the prototype





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Thank you